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Low-temperature liquid-assisted vibrational stress-relief strategy enables stable and efficient perovskite solar cells

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Abstract: The commercialization of perovskite photovoltaics is hindered by intrinsic defects and interfacial strain in polycrystalline films, which compromise both efficiency and stability. Conventional chemical passivation is frequently composition- and structure-dependent and introduces heterogeneous impurities that generate additional stress. Here, we introduce a universal low-temperature liquid-assisted vibrational stress relief (LTL-VSR) strategy that directly addresses the root causes of material degradation through physical acoustic vibration. High-frequency vibrations induce cavitation collapse and resonant lattice sliding, removing nanoscale impurities, annihilating ionic vacancies, and reducing residual stress by over tenfold without thermal damage. This mechanical healing strategy is broadly compatible with diverse perovskite compositions, device architectures, and sizes, delivering power conversion efficiencies (PCEs) of 26.47% for rigid and 24.64% for flexible cells. Moreover, large-area modules achieved PCEs of 23.59% and 20.55% for aperture areas of 16.8 cm² and 600 cm², respectively. Unencapsulated n-i-p devices retain 92.4% of initial efficiency after 1,200 hours of operation. The universality and scalability of LTL-VSR position it as a transformative approach for the industrial manufacturing of perovskite photovoltaics.

Keywords: perovskite solar cells; efficiency and stability; low-temperature liquid-assisted vibrational stress-relief strategy; cavitation mechanism; lattice relaxation; defect elimination

Introduction

Halide perovskites possess exceptional commercial potential for high-efficiency and stable perovskite solar cells (PSCs) owing to their unique octahedral framework and low-cost solution processability¹⁻⁴. However, their path toward commercialization remains hindered by a fundamental limitation: the intrinsic crystalline imperfections of solution-processed polycrystalline films. Generally, variations in crystallization dynamics and non-uniform thermal distribution during film deposition inherently generate vacancy-driven local stress⁵, exaggerated octahedral tilts at grain boundaries^{6,7}, and interfacial thermal-expansion mismatch⁸⁻¹⁰. The resulting heterogeneous stress distribution and its associated localized defects degrade the intrinsic crystalline quality of polycrystalline perovskite films, imposing a critical ceiling on both device efficiency and long-term operational stability¹¹⁻¹³.

Current strategies to mitigate these issues predominantly relied on extrinsic chemical interventions, such as interfacial modification layers or bulk additives^{10,14-18}. Yet these approaches often introduce new sources of instability. For example, typical interface-engineering agents, such as organic ammonium salts used to form 2D/3D heterojunctions, tend to deprotonate under illumination, migrate into the 3D perovskite lattice, and react with formamidinium (FA^+) to form deleterious secondary phases¹⁵. Likewise, although bulk additives can effectively passivate defects in polycrystalline perovskite via specific active sites, their high reactivity under photothermal conditions often yields unpredictable by-products¹⁷. Ultimately, these extrinsic constituents assemble into complex heterostructures that precipitate heterogeneous nanoscale impurities, creating fresh stress concentration sites that severely undermine mechanical integrity and chemical

stability of the perovskite lattice¹⁹⁻²². Moreover, conventional strategies typically enhance the performance of perovskite photovoltaics only for specific compositions and device architectures. Consequently, a universal strategy that intrinsically mitigates underlying structural stress and defects in perovskite films without introducing extrinsic instabilities is urgently needed to achieve efficient and stable PSCs.

Here, we propose a universal low-temperature liquid-assisted vibrational stress-relief (LTL-VSR) strategy that directly addresses this challenge through the cavitation mechanism caused by physical acoustic field without any extrinsic chemical constituents. During the LTL-VSR process, high-frequency mechanical excitation induces liquid-mediated oscillations that drive resonant lattice vibrations, facilitating micro-lattice homogenization and reversible lattice relaxation. Under these vibrational conditions, surface-unstable iodine ions migrate toward internal iodine vacancies, thereby eliminating bulk defects within perovskite films²³⁻²⁵. The collapse of cavitation bubbles generated by the vibrating liquid impart localized shear stress that strips excess PbI_2 from grain boundaries and interfaces. Concurrently, the liquid medium (e.g., anisole) forms Pb-O bonds with under-coordinated Pb^{2+} at grain edges, suppressing PbI_2 re-formation, while the low-temperature protocol prevents vibration-induced thermal damage. This concerted action not only homogenizes perovskite films, eliminating micro-lattice distortions and reducing film stress by more than an order of magnitude, but also eradicates defects such as nanophase impurities and iodide vacancies. We demonstrate the universality and efficacy of LTL-VSR across diverse perovskite compositions, device architectures, and sizes. Treated PSCs exhibit champion power conversion efficiencies of 26.47% (1.54 eV, n-i-p), 26.29% (1.54 eV, p-i-n), 23.27% (1.68 eV, p-i-n), 24.64% (flexible n-i-

p), and 25.37% (1.01 cm^2 , n-i-p). Scaled-up modules further deliver PCEs of 23.59% and 20.55% for aperture areas of 16.8 cm^2 and 600 cm^2 , respectively. Notably, unencapsulated n-i-p devices retain over 92% of their initial efficiency after 1200 h of continuous operation under 1-sun illumination at 65°C , while encapsulated devices maintain 91.7% of their initial efficiency after 1008 h under the ISOS-L-3 protocols. These results establish LTL-VSR as a versatile, scalable, and intrinsically benign processing strategy for producing high-performance and durable perovskite photovoltaics.

Results and Discussion

Dynamics of energy transfer during LTL-VSR

The cornerstone of our LTL-VSR strategy is a precisely engineered system that delivers controlled mechanical energy to promote lattice relaxation and defects elimination without causing structural damage. The finalized apparatus features a high-frequency ultrasonic probe that transmits oscillations through a temperature-controlled liquid medium to the perovskite film (Fig. 1a, Supplementary Fig. 1).

Finite element analysis (FEA) of the gas-liquid exchange under high-frequency vibration reveals the dynamic energy-transfer process (Supplementary Fig. 2). Simulations show that these vibrations induce periodic compression-expansion cycles in the liquid, leading to the nucleation and growth of cavitation bubbles. Subsequent compression triggers bubble collapse, rapidly refilling the residual cavities with liquid and forming shear stress²⁶. Over time, this cyclical process evolves from an initial proliferation of gas phases into a steady gas-liquid exchange equilibrium (Fig. 1b), ensuring consistent and uniform energy delivery to the perovskite film. Fig. 1c illustrates

the interactions between the liquid medium and the perovskite film during the energy-transfer process. Bubbles nucleated under high-frequency vibration fragment into numerous microbubbles upon contacting the film surface^{27,28}. These microbubbles undergo repeated, discrete collapse cycles during successive interactions, penetrating grain boundaries and dissipating energy in a controlled manner.

FEA simulates the gas-liquid distributions across the grain boundaries under varying vibration intensities (Fig. 1d, Supplementary Fig. 3). At insufficient intensity, gas-liquid exchange is weak and bubble collapse is negligible, resulting in inefficient energy transfer. In contrast, excessive intensity promotes gas accumulation and coalescence into microcavities at grain boundaries. The catastrophic collapse of these microcavities instantaneously releases localized energy, fracturing the lattice. At the optimal intensity, stable gas-liquid exchange is sustained along grain boundaries and interfaces, enabling steady energy delivery for stress relaxation and defects elimination.

Design and regulation of the LTL-VSR system

The successful implementation of LTL-VSR strategy critically depends on the rational design of the liquid medium and precise regulation of the vibration source that enables efficient vibration transfer without causing structural damage to the perovskite films.

We first identified the Gutmann donor number (D_N) and dielectric constant (ϵ_r) as the key parameters (Fig. 1e), which together represent the strength of solvent-perovskite polycrystals interactions and the capacity to modulate chemical reactions^{29,30}. Our analysis reveals distinct solvent regimes (Fig. 1f, Supplementary Fig. 4). Solvents with $D_N > 18$ or $\epsilon_r > 30$ exhibit strong

reactivity and readily disrupt the perovskite octahedral structure even in the absence of LTL-VSR treatment. This behavior arises because high- D_N solvents, possessing strong Lewis basicity, coordinate readily with A-site cations (e.g., FA^+), whereas high- ϵ_r solvents facilitate anions dissociation.

In contrast, solvents with $D_N < 5$ and $\epsilon_r < 10$ are largely inert toward perovskite films due to their lack of active coordination sites. Although this inertness preserves lattice integrity, it renders such solvents ineffective for interacting with the perovskite during LTL-VSR. Meanwhile, solvents with $D_N < 18$ and $10 < \epsilon_r < 30$ demonstrate excessive reactivity and solubility toward perovskite polycrystals during LTL-VSR treatment. Once mechanical energy surpasses the activation barrier, irreversible structural degradation occurs.

Guided by this framework, we focused on solvents with intermediate donor numbers ($5 < D_N < 18$) and low dielectric constants ($\epsilon_r < 10$) to achieve a balance between benign interaction and lattice preservation during LTL-VSR. Among the candidates, including ethyl acetate (EA), anisole (Ani), and mesitylene (MSN), EA initially appeared promising. However, its low boiling point (77°C) led to volatilization under localized heating from cavitation, compromising stability during LTL-VSR (Supplementary Fig. 5). By contrast, Ani and MSN, with higher boiling points, maintained stable liquid phases throughout processing.

To probe the specific solvent-perovskite interactions, we conducted spectroscopic analyses. Nuclear magnetic resonance (^1H NMR) spectra of Ani after contact with perovskite films exhibited distinct peak splitting at approximately 3.75 ppm and 7.31 ppm, indicating coordination between its methoxy groups and under-coordinated Pb^{2+} sites on the perovskite surface (Fig. 1g). This

interaction was further confirmed by Fourier-transform infrared (FTIR) spectroscopy (Supplementary Fig. 6). Such weak yet effective Pb-O coordination facilitates passivating unstable dangling bonds and polishing lattice-interface during LTL-VSR. In contrast, MSN showed no evidence of interaction, retaining its characteristic spectral features owing to its symmetric structure and zero dipole moment (Supplementary Fig. 7), confirming its poor coordination ability. Consequently, Ani was identified as the optimal liquid-phase medium for LTL-VSR.

We further engineered the physical properties of the liquid by adding trace amounts of PMMA to tune its dynamic viscosity to an ideal range of 1-3 mPa·s, thereby reducing vibration damping and cavitation threshold (Supplementary Fig. 8) and ensuring efficient vibration transmission and cavitation.^{31,32} The Ani-PMMA solution maintained this ideal viscosity up to 50°C, mitigating the risk of viscosity drop from localized overheating during LTL-VSR, which could otherwise compromise perovskite film quality. To ensure a stable thermal environment, we employed a cooling platform. Without the cooling platform, the system temperature rises sharply by approximately 25°C, from ~25°C to ~50°C, within 9 min of continuous operation. In contrast, when the cooling platform is used under otherwise identical conditions, the temperature increase is limited to only ~7°C over the same period, effectively suppressing thermal degradation of the perovskite films caused by vibrational heating (Fig. 1h, Supplementary Fig. 9).

Finally, we optimized the vibrational energy output, as the vibrational source significantly influences the interaction between the liquid medium and the perovskite film during processing. Overall, the size of the ultrasonic probe directly determines the vibration intensity. A small probe (diameter of 2 mm) provides insufficient intensity to effectively eliminate film defects, whereas a

large probe (6 mm) generates excessive intensity that can readily damage the film. In contrast, a medium-sized probe (3 mm) is expected to enhance film quality without inducing damage (Supplementary Fig. 10). Accordingly, to identify the optimal processing window, we employed a medium-sized probe and systematically investigated the effects of ultrasonic power and treatment duration on film performance. By analyzing X-ray diffraction (XRD) intensity ratio between α -FAPbI₃ and PbI₂ peaks under various vibration powers and durations, a well-defined operational window (80 W and 6 min) was identified (Fig. 1i), achieving maximal benefit without causing damage.

Microstructure and morphology of perovskite films under LTL-VSR

With the system optimized, we confirmed the efficacy of the LTL-VSR strategy in facilitating stress relaxation and defects elimination within the perovskite film by analyzing the evolution of the perovskite crystal lattice during treatment. In-situ XRD conducted during LTL-VSR treatment revealed a pronounced sharpening and an approximately 2.5-fold increase in the intensity of the α -FAPbI₃ (100) peak after 6 minutes. The peak simultaneously shifted from 13.94° to 14.05°, accompanied by the disappearance of PbI₂ features (Fig. 2a, b). These results indicate the removal of lead iodide impurities and a lattice contraction caused by substantial strain relaxation. Over-treatment, however, reversed these improvements, leading to the reappearance of PbI₂ and degradation of the α -FAPbI₃ peak, which signifies the collapse of an overstressed framework. Two-dimensional grazing-incidence wide-angle X-ray scattering (2D GIWAXS) measurements revealed the disappearance of the PbI₂ diffraction ring and a narrowed full width at half maximum (FWHM) for the perovskite phase after 6 minutes, corroborating enhanced crystallographic order

and effective impurities removal (Fig. 2c, d; Supplementary Fig. 11).

X-ray photoelectron spectroscopy (XPS) provided atomic-level evidence of lattice regulation. At the buried interface of untreated films, the Pb $4f_{7/2}$ and Pb $4f_{5/2}$ peaks exhibited shoulder features at 139.37 eV and 144.57 eV, respectively, along with a 0.27 eV shift in binding energy relative to the top surface (Fig. 2e, f). This shift indicates stress-induced elongation of Pb-I bonds and the presence of uncoordinated lead species arising from iodine vacancies. After LTL-VSR treatment, these shoulder features vanished, and the energy offset between the top and buried surfaces became negligible (Fig. 2e, f). The Pb 4f FWHM also became narrower. Similar variation tendency was observed in the I $3d_{5/2}$ and I $3d_{3/2}$ peaks (Supplementary Fig. 12). These results further indicate the release of tensile stress and the filling of vacancy defects, fostering a more uniform local coordination environment around Pb^{2+} and enabling reversible lattice relaxation.

The morphological transformation was examined using scanning electron microscopy (SEM) (Fig. 2g). Untreated films exhibited PbI_2 aggregates at grain boundaries, forming insulating nanosheets that introduced deep-level traps and facilitated crack propagation during photothermal cycling, thereby accelerating film degradation³³. Moreover, mismatches in the coefficients of thermal expansion generated voids on the buried surface of perovskite films, which served as nucleation sites for rapid deactivation. In contrast, LTL-VSR treatment produced a dense, pinhole-free microstructure with impurities effectively removed, verifying the efficacy of the surface polishing process. Moreover, we performed high-resolution transmission electron microscopy (HRTEM) to directly examine the structure and element distribution at grain boundaries (Supplementary Fig. 13). In untreated films, HRTEM clearly reveals 2H-hexagonal phase of PbI_2

at the grain boundaries. After LTL-VSR treatment, neither amorphous PbI_2 nor the 2H-hexagonal PbI_2 phase is observed at stacking faults or grain-boundary intersections in $\alpha\text{-FAPbI}_3$, further confirming the effective elimination of PbI_2 .

This microstructural homogenization and improvement in the quality of film directly translated to superior electronic properties. Kelvin probe force microscopy (KPFM) revealed a dramatic uniformization of the surface potential following LTL-VSR treatment, with the FWHM of the potential distribution narrowing from approximately 39 eV to 19.8 eV (Fig. 2h, i). By contrast, untreated samples exhibited pronounced potential fluctuations at the grain boundaries, which could promote carrier trapping. These results indicate a substantial reduction in electronic disorder and interfacial defect density following treatment, thereby suppressing carrier trapping and promoting efficient charge extraction and transfer (Supplementary Fig. 14).

Defect and stress evolution of perovskite films under LTL-VSR

To further validate that the LTL-VSR strategy effectively removes defects and relieves stress, we directly investigate the evolution of defects and stress in perovskite films during treatment. The defect density was quantitatively evaluated by analyzing the trap density of states (tDOS) through the thermal admittance spectroscopy (TAS) (Fig. 3a, Top and Supplementary Note 1). The tDOS spectra revealed three distinct trap states at approximately 0.24 eV (I), 0.33 eV (II), and 0.45 eV (III). Trap bands I and II correspond to shallow trap (< 0.35 eV) arising from bulk point vacancies and slight distortions caused by octahedral tilting within the lattice, whereas the deep trap band III (> 0.35 eV) is attributed to nanoscale impurities and interfacial strain induced by thermal expansion mismatch^{34,35}. The LTL-VSR-treated devices exhibited a pronounced suppression of defect

densities across all energy levels, indicating a comprehensive enhancement of the intrinsic quality of the films. This reduction was spatially resolved by drive-level capacitance profiling (DLCP) (Fig. 3a, bottom), which also revealed a significant decrease in defect density, especially near the buried interface (550-570 nm). This improvement primarily results from vibration-induced removal of interfacial impurities and vacancy filling. Furthermore, the built-in potential (V_{bi}), derived from Mott-Schottky analysis, increased from 0.97 V to 1.08 V, indicating more efficient photocarrier separation, a result also attributed to the reduction of defects (Supplementary Fig. 15)³⁶.

Moreover, photoluminescence (PL) and time-resolved PL (TRPL) spectra were recorded throughout LTL-VSR treatment. The PL intensity increased during the initial phase, reaching a maximum at 6 minutes before declining with extended treatment (Fig. 3b). This evolution closely mirrors the defect-density trend, with the intensity peak corresponding to the optimal removal of defects and the suppression of non-radiative recombination. TRPL measurements showed an extended carrier lifetime under optimal conditions, corroborating the reduction of non-radiative recombination resulting from decreased defect density (Supplementary Fig. 16 and Supplementary Table 1). Additionally, after 6 minutes of treatment, the PL emission peak exhibited a blue shift from 815 nm to 806 nm, attributed to the contraction of Pb-I bonds that relieve tensile stress. This contraction caused the conduction band minimum to shift downward relative to the valence band maximum, thereby narrowing the bandgap^{37,38}. In contrast, deep trap-tail states at the band edges were observed in the PL spectra, with the PL peak showing both intensity reduction and a red-shift after prolonged resonance treatment. Meanwhile, TRPL measurements revealed a decrease in

carrier lifetime. This is attributed to excessive treatment causing rupture of the octahedral framework, lattice fatigue, and the reformation PbI_2 nanocrystals.

Grazing-incidence XRD (GIXRD) measurements of the $\alpha\text{-FAPbI}_3$ (200) peak revealed a contraction in d-spacing from 3.165 Å to 3.159 Å after 6 minutes of treatment (Supplementary Note 2), also demonstrating tensile-stress release (Fig. 3c). Further $2\theta\text{-sin}^2\psi$ fitting showed a reduction in residual stress from approximately 165 MPa to 16 MPa after treatment, confirming near-complete relaxation of lattice stress during vibrational processing (Fig. 3d, Supplementary Fig. 17). The LTL-VSR strategy is not limited by the thickness of perovskite films and can effectively relieve residual stress of perovskite films across the typical thickness range from 380 to 1000 nm, as detailed in Supplementary Fig. 18.

Defect elimination and stress relief mechanisms

Building on the aforementioned physical framework, we propose a vibration-assisted defect-elimination and stress-relief mechanism for LTL-VSR. As schematic illustration in Fig. 3e, micro-lattice distortions inherent in polycrystalline perovskite films disrupt the stable $[\text{PbX}_6]^{4-}$ coordination, generating both bulk iodide vacancies and surface dangling bonds including uncoordinated lead and iodide ions²⁴. This creates a chemical potential gradient, with a higher energy state at the uncoordinated surface relative to the bulk, thermodynamically driving the migration of surface I^- to fill bulk vacancies^{23,39} (Supplementary Fig. 19). Energy transfer during the LTL-VSR process activates this pathway by inducing resonant lattice vibrations, which promote relative sliding within the octahedral framework and transiently open channels for iodide migration until the bulk vacancies are filled. Meanwhile, the liquid medium (e.g., anisole) interacts

with uncoordinated Pb^{2+} at the surface to form Pb-O bonds. These Pb-O bonds, residual surface dangling bonds, and interfacial PbI_2 nanophase impurities are effectively removed by the localized shear stress generated through cavitation bubble collapse.

Concurrently, the internal stresses are gradually relieved through resonant tensile-compressive cycling of the perovskite lattice during the LTL-VSR treatment, which helps restore optimal bond lengths and angles. Additionally, irregular grain-boundary tilts are alleviated, resulting in a more ordered and stable octahedral framework. FEA further elucidated the stress evolution within film during treatment (Supplementary Fig. 20). Untreated films exhibit a stress gradient that decreases from junctions to grain boundaries and into grain interiors, indicating hierarchical lattice distortion and stress-concentration sites concentrated at junctions and boundaries. During treatment, top-surface stress decreases rapidly and monotonically, with junctional stress dropping most strongly (~ 170 to ~ 14 MPa; Fig. 3f), consistent with GIXRD. Cross-sectional and bottom-surface analyses confirm that stress relaxation propagates uniformly throughout the film bulk (Supplementary Fig. 21), enabled by complete liquid permeation and efficient energy transfer under acoustic vibration. Time-resolved profiles reveal sequential relaxation: junctional stress dissipates first, followed by grain-boundary stress (Supplementary Fig. 22), underscoring the progressive lattice homogenization achieved by LTL-VSR treatment.

To substantiate defect elimination and stress relief, Arrhenius analysis and density functional theory (DFT) were performed (Supplementary Fig. 23). Arrhenius plots revealed that the ion migration activation energy increased from 0.198 eV to 0.323 eV, indicating enhanced lattice homogenization and stabilization resulting from bulk vacancy filling and stress release following

treatment¹¹. DFT simulations computed iodide ion migration energies under different stress states (Fig. 3g). In a tensile-distorted lattice (Pb-I-Pb angle $\sim 163^\circ$, Pb-I length ~ 3.17 Å), iodine vacancies are prone to form, and the ion migration barriers are relatively low: 0.211 eV (out-of-plane) and 0.203 eV (in-plane) (Fig. 3h). After simulated stress release, the lattice adopts a denser, more ordered configuration (Pb-I-Pb angle $\sim 172^\circ$, Pb-I length ~ 3.15 Å), and the corresponding migration energies increase to 0.338 eV and 0.327 eV, respectively, closely matching the experimental values.

Photovoltaic performance and the universality of LTL-VSR

To quantify the photovoltaic impact of LTL-VSR, we therefore fabricated typical n-i-p-structured devices using FAPbI₃ perovskite *via* one-step deposition. After LTL-VSR treatment, the champion device achieved a PCE of 26.47%, with a V_{OC} of 1.188 V, a J_{SC} of 26.23 mA cm⁻², and an FF of 84.94% without any bulk additive or interface modifications (Fig. 4a, Supplementary Fig. 24). This represents a substantial improvement compared with the untreated control device, which delivered a PCE of 24.56%, a V_{OC} of 1.147 V, a J_{SC} of 25.75 mA cm⁻², and an FF of 83.16%. In contrast to the unstable output observed in control devices, the LTL-VSR-treated device demonstrated a stable steady-state power output of 26.06% over 300 s, with no measurable decay. The J_{SC} of 25.88 mA cm⁻² was nearly identical to the J_{SC} integrated from external quantum efficiency (EQE) measurement (25.83 mA cm⁻²), agreeing well with the results of J - V curves (Fig. 4b, Supplementary Fig. 25). Statistical analysis of 20 independent devices clearly confirms the reproducibility of the observed performance enhancement. (Fig. 4c).

Crucially, a key advantage of the LTL-VSR strategy is its broad universality, including

applicability across diverse perovskite compositions and device architectures (Supplementary Table 2), as well as compatibility with conventional chemical passivation strategies, flexible processing, and large-scale fabrication. For instance, in a wider-bandgap (1.68 eV) composition, $\text{Rb}_{0.05}\text{Cs}_{0.05}\text{MA}_{0.15}\text{FA}_{0.75}\text{Pb}(\text{I}_{0.76}\text{Br}_{0.24})_3$, LTL-VSR boosted the PCE to 22.53%, surpassing the control device's PCE of 20.50% (Supplementary Fig. 26). For p-i-n structured devices using $\text{Cs}_{0.05}\text{FA}_{0.95}\text{PbI}_3$, the PCE increased from 24.47% to 26.29% after LTL-VSR treatment (Supplementary Fig. 27). To further demonstrate compatibility with chemical passivation, we employed a representative state-of-the-art interfacial passivation strategy using 2-phenylethylammonium iodide (PEAI) to form a 2D/3D heterojunction. When combined with LTL-VSR, the p-i-n devices achieved an enhanced PCE of 26.41% (Supplementary Fig. 28), confirming the effective integration and synergistic potential of the two approaches. This enhancement can be attributed to the LTL-VSR treatment first relieving intrinsic residual stress and defects, thereby providing a more uniform and stable foundation for subsequent chemical passivation. The passivation layer further reduces surface defects, while the lattice relaxation induced by LTL-VSR suppresses organic molecule migration and stress accumulation, resulting in a synergistic improvement. The strategy also proved highly effective for flexible photovoltaics (Supplementary Fig. 29), enhancing the PCE of f-PSCs from 22.65% to 24.64%, with a V_{oc} of 1.148 V, a J_{sc} of 25.66 mA cm^{-2} , and an FF of 83.65%. Notably, LTL-VSR is readily scalable. When the active area was increased to 1.01 cm^2 , the PCE improved from 23.20% to 25.37% following treatment (Supplementary Fig. 30). To verify the effect of the LTL-VSR strategy on the uniformity of large-area perovskite films, $5 \times 5 \text{ cm}^2$ films were prepared and subjected to grid-based multi-point XRD

characterization with 25 measurement points per film (Supplementary Fig. 31). The relative standard deviation (RSD) of the α -FAPbI₃(100)/PbI₂(100) peak ratio was 3.43%, with no notable difference between the center and the edge, confirming the compatibility of this strategy with large-area films. Correspondingly, the 5×5 cm² module (aperture area 16.8 cm²) achieved a PCE of 23.59% (Fig. 4d and Supplementary Table 4). Moreover, 30×30 cm² modules (aperture area 600 cm²) were fabricated to validate scalability, yielding a reverse-scan PCE of 20.55% (Fig. 4e and Supplementary Table 5). These results indicate that the high-frequency vibrations transmitted through the liquid-phase medium enable uniform treatment of the entire perovskite film and demonstrating the scalability of this region-independent strategy.

Beyond efficiency, operational stability is the paramount metric for perovskite photovoltaics. We subjected unencapsulated n-i-p devices to the International Summit on Organic Photovoltaic Stability (ISOS) protocols. Under continuous thermal stress (ISOS-D-2, 65°C in N₂), LTL-VSR-treated devices retained approximately 94% of their initial PCE after 2000 h, significantly outperforming control devices, which degraded to around 73% (Fig. 4f). Furthermore, under MPP (maximum power point) tracking at 65°C and 1-sun illumination (ISOS-L-2), the treated devices maintained 92.4% of their initial efficiency after 1200 h, while the controls rapidly degraded to below 60% (Fig. 4g). Even more impressively, we investigated the storage stability of encapsulated n-i-p PSCs under ISOS-D-3 conditions (85°C, 85% RH). The LTL-VSR-processed n-i-p devices retained 93.1% of their initial performance after 1008 h, whereas the control devices retained less than 70% after 504 h. MPP tracking under harsh ISOS-L-3 conditions (85°C, 85% RH) further confirmed the superior operational durability of the treated devices. Specifically, the modified n-

i-p devices maintained 91.7% of their initial efficiency after 1008 h, in stark contrast to the control devices, which rapidly degraded to 47.8% after 500 h (Supplementary Fig. 32). The performance comparison of LTL-VSR-treated devices with previously reported approaches is summarized in Supplementary Table 3. This exceptional performance arises from the fundamental improvements imparted by LTL-VSR, namely a more mechanically robust lattice due to stress release and the elimination of defect that typically serve as nucleation sites for degradation.

In summary, we have introduced a low-temperature, liquid-assisted vibrational stress relief strategy as a transformative treatment paradigm for perovskite photovoltaics. This approach uniquely enables the intrinsic enhancement of polycrystalline films, simultaneously eliminating defects and releasing stress without resorting to extrinsic chemical additives. We have elucidated the underlying mechanism in which promotes ion diffusion and vacancy filling, leading to defect annihilation, while energy released from controlled cavitation bubble collapse provides a distinct energy-transfer pathway for microstrain homogenization. The exceptional material quality is directly evidenced by a dramatic reduction in trap state density and a near-complete release of residual tensile stress. Consequently, we demonstrated high-performance and operationally stable solar cells across multiple perovskite architectures and compositions, underscoring the considerable potential of LTL-VSR for scalable industrial manufacturing. This work establishes a foundational mechanical strategy for lattice regulation, opening a distinct avenue beyond molecular passivation for the future of high-performance and durable optoelectronics.

Methods

Materials

The transparent fluorine-doped tin oxide (FTO) glass, indium tin oxide (ITO) glass, PEN/ITO, nano-nickel oxide (NiO_x) powder were purchased from Advanced Election Technology Co., Ltd. Lead iodide (PbI₂, 99.99%), Lead bromide (PbBr₂, 99.99%), cesium iodide (CsI, 99%), Silver(I) bis(trifluoromethanesulfonyl)imide (Ag-TFSI, 98%), 4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl] phosphonic acid (Me-4PACz, 99%) was purchased from TCI. Formamidinium iodide (FAI, ≥99.5%), methylammonium bromide (MABr, ≥99.5%), methylammonium chloride (MACl, ≥99.5%) were purchased from Greatcell. Rubidium iodide (RbI), tin chloride dihydrate (SnCl₂·2H₂O, 99.99%), urea, thioglycolic acid (TGA, 98%), Bathocuproine (BCP, 96%) were purchased from Sigma-Aldrich. 2,2',7,7'-tetrakis(*N,N*-di-*p*-methoxyphenylamine)-9,9'-spirobifluorene (Spiro-OMeTAD) and [6,6] phenyl-C61-butyric acid methyl ester (PC₆₁BM, 99.8%) were Luminescence Technology Corp. *N*-Propyl-3-methylpyridinium bis(trifluoromethylsulfonyl)imide (PMPy-TFSI, 99%), 1,3,5-Trimethyl benzene (MSN, ≥99%), anisole (Ani, ≥99.7%) were purchased from Aladdin. The commonly used solvents including *N,N*-dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, ≥99.9%), ethyl acetate (EA, ≥99.5%) were purchased from Sigma-Aldrich. Diethyl ether (DE, 99.7%) and hydrochloric acid (HCl, 37 wt. % in water) were purchased from Shanghai Lingfeng Chemical Reagent Co., Ltd. α-FAPbI₃ was synthesized by the inverse temperature crystallization method⁴⁰. All chemical reagents were used as supplied without any additional purification unless otherwise specified.

Preparation of perovskite films

FAPbI₃ perovskite for n-i-p-structured devices: the perovskite precursor solution was prepared by mixing 1.82 M PbI₂, 1.7 M FAI, 0.6 M MACl in mixed DMF/DMSO solvent with volume ratio 8:

1. The solution was vortexed for 2 hours to ensure complete dissolution, after which it was transferred into glove box and cooled to 0°C for subsequent use. The perovskite precursor was spin-coated on the as-prepared substrates at 1000 rpm for 10 s, subsequently at 5000 rpm for 30 s, during which 700 μ L DE was dripped onto the center of film at 15 s before the end of spin-coating. The substrates were then annealed at 120°C for 60 min. It is noteworthy that both the flexible and 1.01 cm² devices were fabricated using the same preparation method.

Cs_{0.05}FA_{0.95}PbI₃ perovskite for p-i-n-structured devices: the perovskite precursor solution was prepared by mixing 1.58 M PbI₂, 0.073 M CsI, 1.38 M FAI, 0.25 M MACl in mixed DMF/DMSO solvent with volume ratio 8: 1. The solution was vortexed for 2 hours to ensure complete dissolution, after which it was transferred into glove box and cooled to 0°C for subsequent use. The perovskite precursor was spin-coated on the as-prepared substrates at 1000 rpm for 10 s, subsequently at 5000 rpm for 30 s, during which 200 μ L EA was dripped onto the center of film at 10 s before the end of spin-coating. The substrates were then annealed at 100°C for 30 min.

Rb_{0.05}Cs_{0.05}MA_{0.15}FA_{0.75}Pb(I_{0.76}Br_{0.24})₃ perovskite for p-i-n-structured devices with bandgap of 1.68 eV: CsI(18.2 mg), RbI(14.8 mg), MABr(23.5 mg), FAI(180.6 mg), PbI₂(507.6 mg, 10% of excess) and PbBr₂(146.4 mg) were dissolved in 1 mL mixed solvent of DMF:DMSO (v: v = 4:1), shaken using a vortex shaker for 2 h to ensure complete dissolution. Moreover, 13 mol% of MACl was added to the perovskite precursor solution to enhance the crystallization. The perovskite precursor was spin-coated on the as-prepared substrates at 1,000 rpm for 10 s, subsequently at 5,000 rpm for 30 s, during which 200 μ L EA was dripped onto the center of film at 10 s before the end of spin-coating. The substrates were then annealed at 100°C for 15 min.

LTL-VSR strategy for perovskite films

The as-deposited perovskite substrates were secured in custom-made fixture coupled to ultrasonic probe and immersed in an Ani-PMMA mixture (0.05 mg mL^{-1} PMMA) resting on cooling platform set to 0°C . Vibration fields of controlled intensity were generated with interchangeable probes ($\varnothing$ 2, 3 or 6 mm) operating at 20-25 kHz (auto-tracked) and 10-100 W output power for 0-10 min. Subsequently, residual surface solvents were removed using gentle nitrogen stream, and the films were transferred into glove box for annealing at 100°C for 5 min.

Fabrication of PSCs

N-i-p-structured devices: The FTO glass substrates were sequentially ultrasonically cleaned in deionized water, acetone and ethanol for 15 min each, followed by ultraviolet ozone (UVO) treatment for 20 min. SnO_2 electron transport layers were deposited via chemical bath deposition (CBD) according to previously reported method^{40,41}. Briefly, substrates were immersed in CBD solution prepared by mixing $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ (108 mg), urea (500 mg), TGA (10 μL), and HCl (500 μL) in 40 ml of deionized water, followed by heating at 90°C for 3 h. The substrates were then annealed at 150°C for 1 h. Notably, the electron transport layer for flexible devices was prepared by spin-coating 2.5 wt% SnO_2 dispersion at 4000 rpm for 30 s onto UVO-treated PEN/ITO substrates. The substrates were then annealed at 150°C for 30 min. After perovskite deposition and subsequent LTL-VSR post-treatment, Spiro-OMeTAD solution (90 mg Spiro-OMeTAD in chlorobenzene with 9.0 mg $\text{spiro}^{2+}(\text{TFSI})_2$ and 3.2 μL PMPy-TFSI) was spin-coated onto the film at 3000 rpm for 30 s. The $\text{spiro}^{2+}(\text{TFSI})_2$ dopant was synthesized following previously reported method⁴², and the solution was filtered through a 0.22 μm PTFE filter prior to use. Finally, ~ 100

nm Ag electrodes were deposited via thermal vacuum evaporation ($< 8 \times 10^{-4}$ Pa).

P-i-n-structured devices: The ITO glass substrates were sequentially ultrasonically cleaned in deionized water, acetone and ethanol for 15 min each, followed by ultraviolet ozone (UVO) treatment for 20 min. $\text{NiO}_x/\text{Me-4PACz}$ was employed as the hole transport layer. NiO_x nanoparticles were dispersed in deionized water at a concentration of 10 mg mL^{-1} to form NiO_x ink. A self-assembled molecular layer solution was prepared by dissolving 0.5 mg mL^{-1} Me-4PACz in ethanol. After UVO treatment, the NiO_x ink was spin-coated onto the substrates at 4000 rpm for 30 s and then annealed at 100°C for 10 min. The substrates were subsequently transferred to a nitrogen-filled glove box. Following this, the Me-4PACz self-assembled molecular solution was spin-coated onto the substrate at 4000 rpm for 30 s and annealed at 100°C for 10 min. After perovskite deposition and subsequent LTL-VSR post-treatment, PC_{61}BM layer (20 mg mL^{-1} in CB) was applied via spin-coating at 3000 rpm for 30 s. BCP solution (0.5 mg mL^{-1} in isopropanol) was then spin-coated onto the PC_{61}BM layer at 5000 rpm for 30 s. Finally, Ag ($\sim 100 \text{ nm}$) was evaporated onto the as-prepared films via thermal vacuum evaporation ($< 8 \times 10^{-4}$ Pa).

$5 \times 5 \text{ cm}^2$ modules: $5 \times 5 \text{ cm}^2$ modules were fabricated using an architecture analogous to that of the n-i-p devices. The PSC modules, with an aperture area of 16.80 cm^2 and comprising eight sub-cells connected in series, were fabricated on $5 \times 5 \text{ cm}^2$ FTO glass substrates. Series interconnection was achieved through P1, P2, and P3 patterning. The gap between adjacent FTO patterns (P1) was $30 \mu\text{m}$, after which the patterned substrates were cleaned. The electron-transport layer, perovskite layer, and hole-transport layer were then deposited following the same procedures described above. For the P2 scribing process, a 532 nm laser (GHLaser) was used, producing a P2

width of 140 μm . Finally, a 100 nm Cu electrode was thermally evaporated under a pressure of 1×10^{-4} mbar, followed by formation of the 90 μm P3 line.

30×30 cm² modules: The PSC modules, with an aperture area of 600 cm² and comprising 45 sub-cells connected in series, were fabricated on 30×30 cm² FTO glass substrates. Series interconnections were implemented through P1, P2, and P3 lines. The gap between adjacent FTO patterns, defined as P1, had a width of 50 μm , and the patterned substrates were subsequently cleaned. Then, the electron transport layers, perovskite and hole transport layers were prepared using blade coating method. The perovskite films were fabricated with vacuum flash evaporation technology. The other treatment steps are the same as presented above. For P2 etching process, the laser used was a 532 nm laser. The width of P2 is 41 μm . A 100 nm Cu electrode was thermally evaporated under 1×10^{-4} mbar, followed by formation of the 50 μm P3 line. The distances between the P1 and P2 scribes and between the P2 and P3 scribes were 40 μm . The geometric fill factor is 96%.

Characterization

Device characterization: The J - V curves were measured using a Keithley 2400 source meter under standard one sun illumination (AM 1.5G, 100 mW cm⁻²) with a voltage step of 0.02 V, provided by a Xenon lamp solar simulator (EnliTech SS-F5-3A). The light intensity was calibrated using a NIST-certified silicon solar cell (Newport 532 ISO1599). The aperture area of each device was defined by a precision shadow mask, yielding 0.05 cm² for regular solar cells and 1.01 cm² for large-area architectures. The EQE spectra of the PSCs was measured on QE-R3018 solar cell spectral response measurement system (Enlitech). The TAS and DLCP measurements were

performed using Wayne Kerr 6500b precision analyzer³⁴. Mott-Schottky analysis was performed over 0-1.2 V at 10 kHz under 20 mV AC bias. The stability of the device was evaluated at the maximum power point under AM 1.5G illumination (100 mW cm^{-2}) provided by an LED lamp (SS150A, Zolix) within a N_2 -filled glovebox. The unencapsulated devices were heated using a hotplate set to 65°C .

Material and film characterization: ^1H -NMR spectra were measured with Bruker AVANCE NEO 400M instrument at 400 MHz and 298 K. FTIR spectrum was obtained using infrared spectrometer (PerkinElmer, Spectrum Two). Rheological characterization was performed using rotary rheometer (TA Discovery, HR-1). The temperature of the liquid phase medium was monitored by thermal imager (IRay, C200 Pro⁺). XRD patterns were recorded using a Bruker D8 Advance A25 under $\text{Cu K}\alpha$ radiation source ($\lambda = 1.54 \text{ \AA}$) with an operating voltage of 40 kV and a current of 40 mA. XRD diffraction patterns at different incident angles were obtained using GIXRD (Bruker D8). GIWAXS measurements were conducted at beamline BL14B1 of the Shanghai Synchrotron Radiation Facility (SSRF). The sample was exposed to an X-ray beam with a wavelength of $0.68877 \text{ \AA}$ at an incident angle of 0.50° , and the resulting scattered radiation was captured using MarCCD detector positioned 746 mm vertically above the sample. KPFM and AFM images were collected using Bruker Dimension Icon system. Surface and cross-sectional morphology of perovskite were performed by field-emission scanning electron microscope (Hitachi S-4800) with a voltage of 5 kV. The XPS characterization of films was characterized by X-ray photoelectron spectroscopy (Shimadzu, Axis Supra). Steady-state PL spectra were recorded on Hitachi F-4600 fluorescence spectrophotometer.

Theoretical calculation and finite element analysis

Our density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package (VASP), employing plane-wave basis sets combined with the projector augmented-wave (PAW) method. Throughout the structural optimization, all atomic configurations were thoroughly relaxed until reaching a state of minimized total energy. The migration energy barrier (ΔE) was determined as the energy difference between the saddle point (highest-energy transition state) and the initial local minimum, representing the activation energy for vacancy migration.

The ground-state geometries of several liquid-phase medium were calculated using Gaussian16W with the B3LYP functional at the 6-31G (d, p) basis set, and the electrostatic potential distributions were subsequently analyzed in conjunction with Multiwfn software⁴³. Polycrystalline film model was constructed, with distinct initial stress distributions assigned at the junctions, grain boundaries, and within the grains⁸. Harmonic displacement excitation was applied to the coupled system to investigate the process of stress relaxation in the thin film. Additionally, a volume of fluid (VOF) model was developed, wherein harmonic displacement served as the excitation source to simulate the gas-liquid exchange process occurring within the polycrystalline model.

Data availability

The data supporting the findings of this study are available within the article and its Supplementary Information. Source data underlying the figures and analyses are provided with this paper.

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Author contributions

W.Y.L. and T.C. conceived the idea and designed the experiments. S.Y. carried out all experimental work. W.W. assisted with the ^1H NMR characterization and DFT calculations. P.Z. and J.G. contributed to the device stability characterization. Z.Y. and C.Z. assisted in preparing the schematic illustrations. S.Y., T.C., Y.C., and W.Y.L. wrote and revised the manuscript. W.Y.L. led the project. All authors discussed the results and contributed to the revision of the manuscript.

Competing interests

The authors declare no competing interests.

Figure legends/captions

Fig. 1. Design and regulation of the LTL-VSR system. **a**, Diagram of the LTL-VSR strategy. TCO and CTL denote transparent conductive oxide and charge transport layer, respectively. **b**, Evolution of gas-liquid ratio at different step. **c**, Energy transfer mechanism of perovskite films under LTL-VSR. **d**, LTL-VSR-driven liquid dynamic simulation with various vibration intensities. Scale bars, 10 nm. **e**, Gutmann donor number (D_N) and dielectric constant (ϵ_r) of various solvents. Tol: toluene; CB: chlorobenzene; DCM: dichloromethane; m-Xyl: m-xylene; PhCN: benzonitrile; BzCl: benzoyl chloride; AcCl: acetyl chloride; AC: acetone; ACN: acetonitrile; NME: nitromethane; DOL: 1,3-dioxolane; THF: tetrahydrofuran; IPA: isopropanol; MeOH: methanol; EtOH: ethanol. **f**, Photographs of 50 mg mL⁻¹ FAPbI₃ dissolved in various solvents for 24 hours with/without LTL-VSR. **g**, ¹H NMR spectra of Ani solution with/without LTL-VSR. **h**, Time-resolved temperature curves with/without cooling platform. Error bars represent the standard deviation of five independent measurements. **i**, Time- and power-dependent XRD peak ratio (FAPbI₃/PbI₂) under LTL-VSR.

Fig. 2. Microstructure and morphology of perovskite films under LTL-VSR. **a-b**, Time-resolved evolution of FAPbI₃ and PbI₂ XRD peaks during LTL-VSR. **c-d**, 2D GIWAXS images of perovskite films without (c) and with (d) LTL-VSR. **e-f**, X-ray photoelectron spectra (XPS) of Pb 4f peaks without (e) and with (f) LTL-VSR. **g**, SEM images of perovskite films with/without LTL-VSR. Scale bars, 500 nm, 2 μ m, respectively. **h**, KPFM images of perovskite films with/without LTL-VSR. Scale bars, 500 nm. **i**, Surface potential distribution of perovskite films with/without LTL-VSR.

Fig. 3. Defect evolution and stress analysis of perovskite films. **a**, tDOS spectra, and DLCP profiles of PSCs with/without LTL-VSR. **b**, PL spectra under LTL-VSR. **c**, In-situ XRD patterns and d-space values of (200) peaks of perovskite films under LTL-VSR. **d**, Linear 2θ - $\sin^2\Psi$ fits of perovskite films with/without LTL-VSR. **e**, Vibration-assisted defect-elimination and stress-relief mechanism for LTL-VSR. **f**, Top and cross-sectional views of time-dependent stress simulation in polycrystalline perovskite films. Scale bars, 2 μm . **g**, Theoretical simulation of migration barriers and lattice variations. **h**, Relative energy variations of diverse theoretical models.

Fig. 4. Photovoltaic performance and stability. **a**, J - V curves of n-i-p PSCs with/without LTL-VSR. **b**, Stabilized photocurrent and power output of PSCs. **c**, Photovoltaic statistics of 20 PSCs with/without LTL-VSR. **d**, J - V curves of $5\times 5\text{ cm}^2$ module based on LTL-VSR. Inset: photograph of $5\times 5\text{ cm}^2$ module. **e**, J - V curves of $30\times 30\text{ cm}^2$ module based on LTL-VSR. Inset: photograph of $30\times 30\text{ cm}^2$ module. **f**, Thermal stability test of unencapsulated PSCs under dark based on 5 devices. **g**, Operational stability test by tracking MPP at 65°C under continuous 1-sun illumination.

Figures

Fig. 1

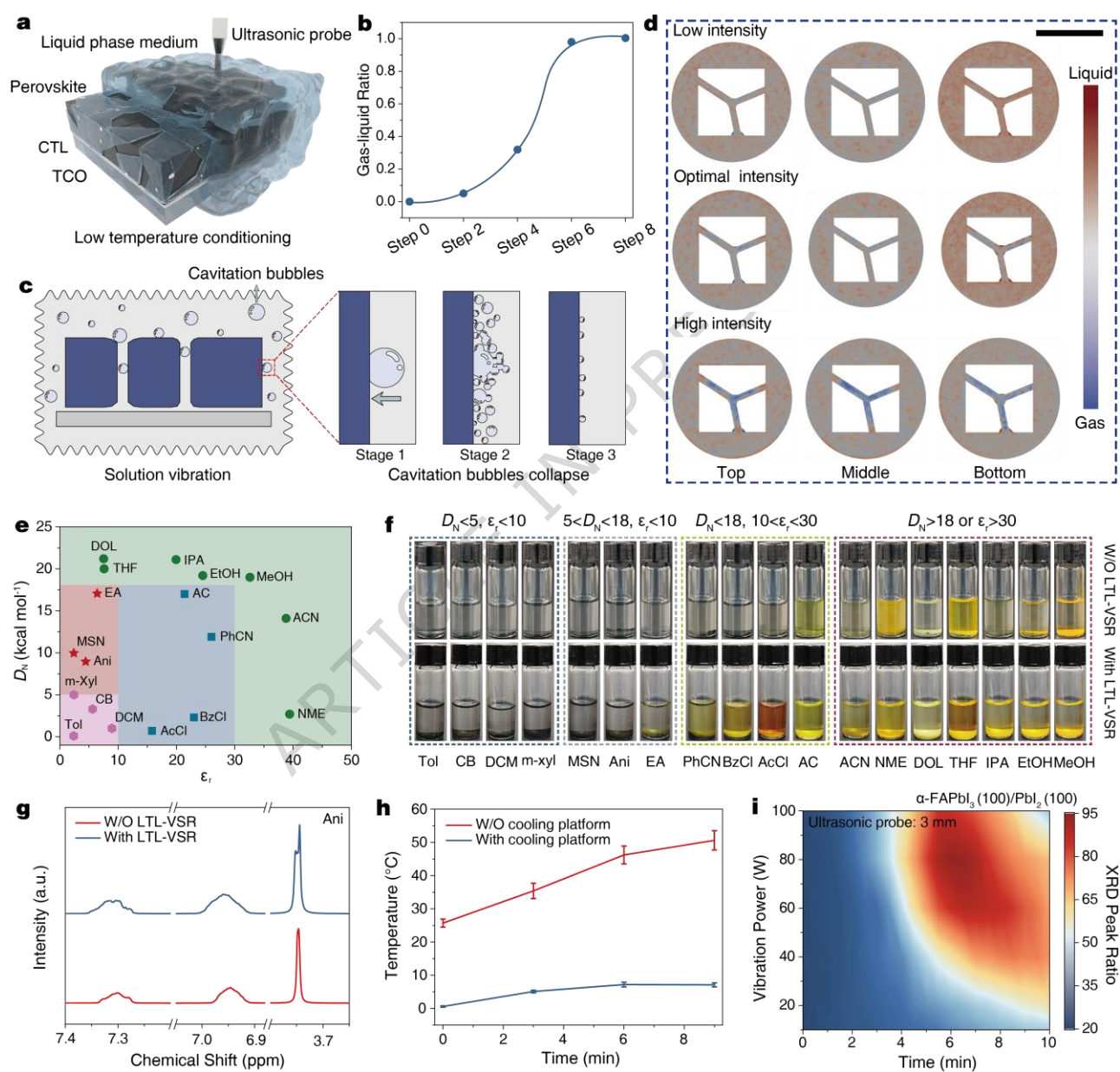


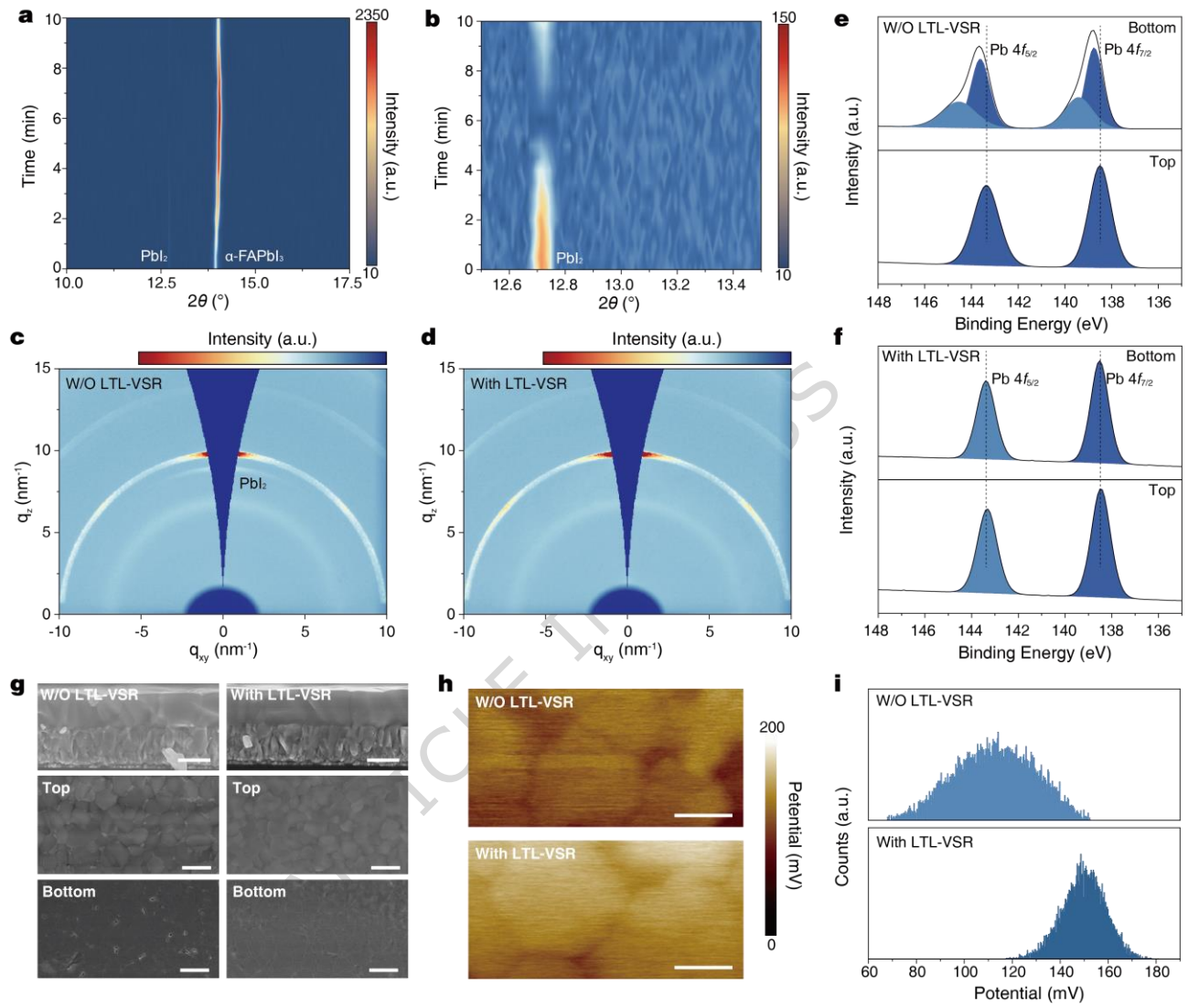
Fig. 2

Fig. 3

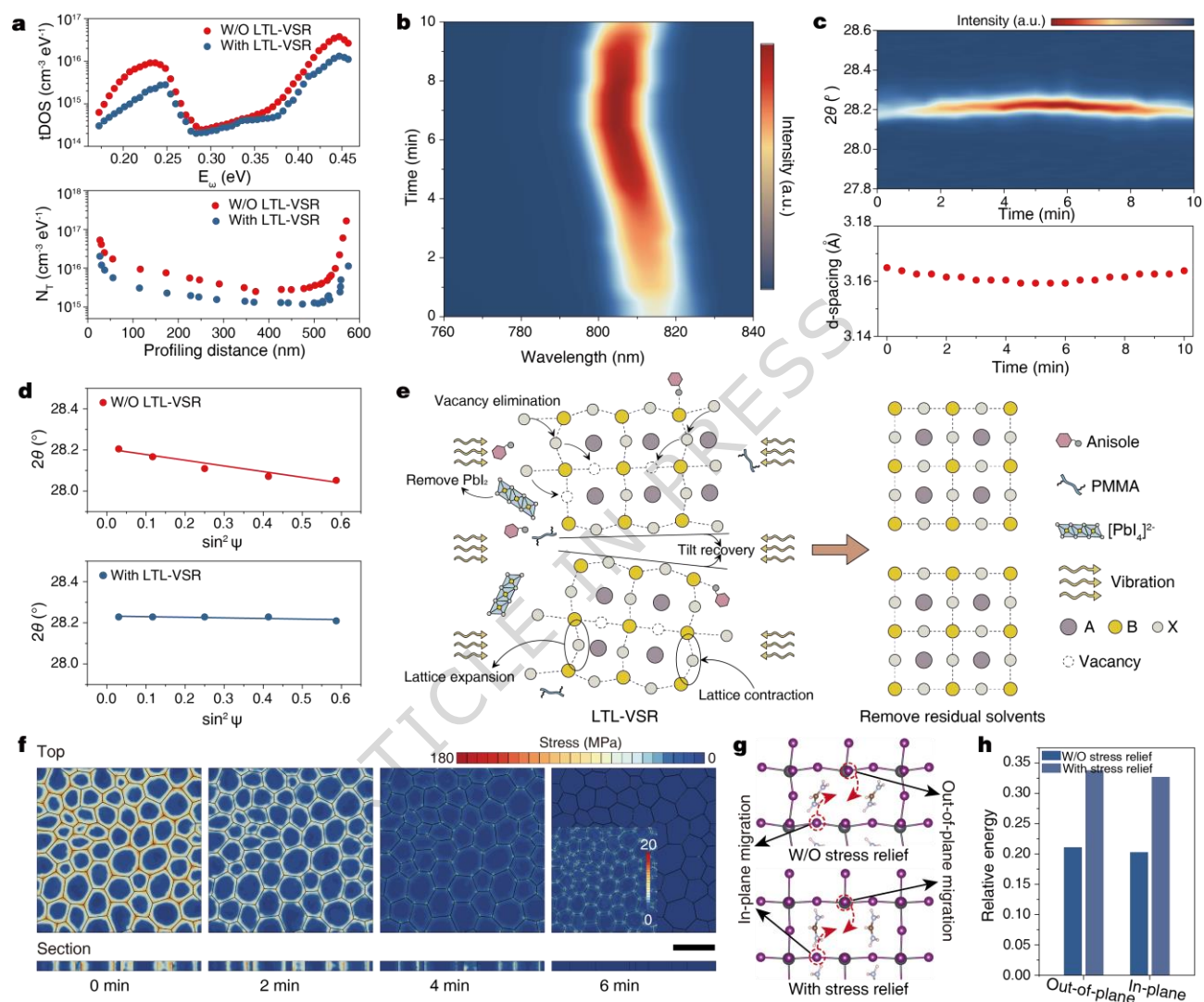


Fig. 4

